

$$\Rightarrow \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix} \text{ (By defn. of } Ax \text{)}$$

By Theorem 3, the above equation has the same solution set as the system of equations corresponding to the augmented matrix $\begin{bmatrix} 1 & 1 & 3 \\ 4 & 5 & 8 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & 3 \\ 4 & 5 & 8 \end{bmatrix} \xrightarrow[\text{-4} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & -4 \end{bmatrix} \xrightarrow[\text{-Row 2}]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & -4 \end{bmatrix}$$

$$\therefore x_1 = 7 \text{ and } x_2 = -4$$

$$T(x_1, x_2) = \begin{bmatrix} x_1 + x_2 \\ 4x_1 + 5x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

$$x = \begin{bmatrix} 7 \\ -4 \end{bmatrix}$$

Note: We must give a rigorous proof that the solution set is unique.

22. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation such that $T(x_1, x_2) = (x_1 - 2x_2, -x_1 + 3x_2, 3x_1 - 2x_2)$. Find x such that $T(x) = (-1, 4, 9)$.

Solution: $T(x_1, x_2) = \begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 - 2x_2 \\ -x_1 + 3x_2 \\ 3x_1 - 2x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ -x_1 \\ 3x_1 \end{bmatrix} + \begin{bmatrix} -2x_2 \\ 3x_2 \\ -2x_2 \end{bmatrix}$

$$= \begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix} \text{ (Vector addition)} \Rightarrow x_1 \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix} \text{ (Scalar multiplication)}$$

$$\Rightarrow \begin{bmatrix} 1 & -2 \\ -1 & 3 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix} \text{ (By defn. of Matrix equation } Ax=b \text{)}$$

By Theorem 3, the above equation has the same solution set as the system of equations corresponding to the augmented matrix $\begin{bmatrix} 1 & -2 & -1 \\ -1 & 3 & 4 \\ 3 & -2 & 9 \end{bmatrix}$

$$\begin{bmatrix} 1 & -2 & -1 \\ -1 & 3 & 4 \\ 3 & -2 & 9 \end{bmatrix} \xrightarrow[\text{+Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 3 \\ 3 & -2 & 9 \end{bmatrix} \xrightarrow[\text{-3} \times \text{Row 1}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 3 \\ 0 & 4 & 12 \end{bmatrix}$$

$$\xrightarrow[\text{-4} \times \text{Row 2}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{+2} \times \text{Row 2}]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$